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4.1.3 What you will learn

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This week is all about solving linear least squares, a fundamental problem encountered when fitting data or approximating matrices.

Upon completion of this week, you should be able to

- Formulate a linear least squares problem.
- Transform the least squares problem into normal equations.
- Relate the solution of the linear least squares problem to the four fundamental spaces.
- Describe the four fundamental spaces of a matrix using its singular value decomposition.
- Solve the solution of the linear least squares problem via Normal Equations, the Singular Value Decomposition, and the QR decomposition.
- Compare and contrast the accuracy and cost of the different approaches for solving the linear least squares problem.

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